F78PA Probability & Statistics A, December 2022/23 Solutions

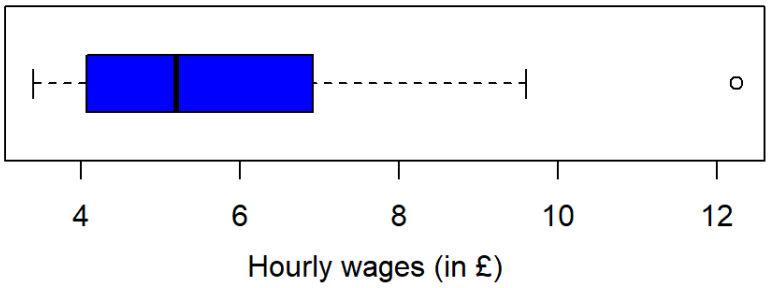
Maps to CS1 Actuarial Statistics exemption.
Weighting within CS1 exemption: 25%.

1. *Similar examples to all parts seen.*
CS1 syllabus objective 2.2.2.

(a) Sample size $n = 15$, so

$$\begin{aligned} \text{median} &= x_{\left(\frac{n+1}{2}\right)} = x_{(8)} = 5.20 \ (£) && \text{[1 mark]} \\ Q_1 = \text{lower quartile} &= x_{(1+0.25\times(n-1))} = x_{(4.5)} = \frac{3.95 + 4.20}{2} = 4.075 \ (£) \\ Q_3 = \text{upper quartile} &= x_{(1+0.75\times(n-1))} = x_{(11.5)} = \frac{6.60 + 7.25}{2} = 6.925 \ (£) \\ IQR = \text{inter-quartile range} &= 6.925 - 4.075 = 2.85 \ (£) && \text{[2 marks]} \end{aligned}$$

(b) Boxplot:



[2 marks]

Comment: the shape of the distribution is skewed to the right with an outlier of £12.25. [1 mark]

(c) Mean $\bar{x} = \frac{90.0}{15} = £6.0$ [1 mark]

Standard deviation $s = \sqrt{\frac{1}{14}\left(631.25 - \frac{90.0^2}{15}\right)} = £2.553$ [2 marks]

(d) Under scheme (I), new sample mean and standard deviation are $\bar{x}_1 = \bar{x} + 0.30 = £6.30$, $s_1 = s = £2.553$

Under scheme (II), new sample mean and standard deviation are $\bar{x}_2 = (1 + 0.05)\bar{x} = £6.30$, $s_2 = 1.05s = £2.681$ [2 marks]

Mean is the same under scheme (I) and scheme (II), so total wage bill is the same under the two schemes, and management wouldn't mind which scheme is introduced.

Standard deviation is greater under scheme (II) than under scheme (I), so some workers benefit more than others, in particular lower paid workers benefit least, so workers may mind which scheme is implemented. [2 marks]

[Total 13 Marks]

2. *Similar examples to all parts seen.*
CS1 syllabus objectives 1.1.1, 1.1.3.

Define events

$$\begin{aligned} A &= \text{component produced in Aberdeen,} \\ B &= \text{component produced in Belfast,} \\ C &= \text{component produced in Cardiff,} \\ D &= \text{component defective.} \end{aligned}$$

Then we are told that

$$\begin{gathered} \Pr(A) = 0.3, \quad \Pr(B) = 0.3, \quad \Pr(C) = 0.4, \\ \Pr(D \mid A) = 0.01, \quad \Pr(D \mid B) = 0.03, \quad \Pr(D \mid C) = 0.02 \end{gathered}$$

(a) $\Pr(D) = \Pr(A)\Pr(D \mid A) + \Pr(B)\Pr(D \mid B) + \Pr(C)\Pr(D \mid C) = 0.3 \times 0.01 + 0.3 \times 0.03 + 0.4 \times 0.02 = 0.02$ [2 marks]

$\Pr(A \mid D) = \frac{\Pr(A)\Pr(D|A)}{\Pr(D)} = \frac{0.3 \times 0.01}{0.02} = \frac{3}{20} = 0.15$ [2 marks]

(b) To justify treating X as binomially distributed, need to assume that whether or not a particular component is defective is independent of whether or not any other components are defective, and that each component has the same probability of being defective. [2 marks]

Under the above assumptions, then from part (a), $X \sim \text{Bin}(20, 0.02)$.
$\Pr(X > 2) = 1 - \Pr(X \leq 2) = 1 - \binom{20}{0}0.02^0 0.98^{20} - \binom{20}{1}0.02^1 0.98^{19} - \binom{20}{2}0.02^2 0.98^{18} = 1 - 0.667608 - 0.272493 - 0.052830 = 0.00707$ [3 marks]

[Total 9 Marks]

3. *Part (a) bookwork; part (b) similar to seen examples.*
CS1 syllabus objective 1.1.2.

(a) For $t < 0$, density is zero, and so $F_T(t) = 0$ for $t < 0$.
For $t \geq 0$,

$$\begin{aligned} F_T(t) = \Pr(T \leq t) &= \int_{-\infty}^{t} f_T(u)\,du = \int_0^t \lambda \mathrm{e}^{-\lambda u}\,du \\ &= \left[-\mathrm{e}^{-\lambda u}\right]_0^t \\ &= 1 - \mathrm{e}^{-\lambda t} \end{aligned}$$

That is,

$$F_T(t) = \begin{cases} 0 & t < 0, \\ 1 - \mathrm{e}^{-\lambda t} & t \geq 0. \end{cases}$$

[3 marks]

Expectation is

$$\begin{aligned} E(T) = \int_{-\infty}^{\infty} t f_T(t)\,dt &= \int_0^{\infty} t\lambda \mathrm{e}^{-\lambda t}\,dt \\ &= \left[-t\mathrm{e}^{-\lambda t}\right]_0^{\infty} + \int_0^{\infty} \mathrm{e}^{-\lambda t}\,dt \text{ (integrating by parts)} \\ &= 0 + \left[-\frac{\mathrm{e}^{-\lambda t}}{\lambda}\right]_0^{\infty} \\ &= \frac{1}{\lambda} \end{aligned}$$

[2 marks]

(b) i. Have $\lambda = 2$ and $Y = 1 - \mathrm{e}^{-T}$. Since $\mathrm{Im}T = [0, \infty)$, then $\mathrm{Im}Y = [0, 1)$. So for $y < 0$, we have $F_Y(y) = 0$, and for $y \geq 1$, we have $F_Y(y) = 1$. [1 mark]
For $0 \leq y < 1$,

$$\begin{aligned} F_Y(y) = \Pr(Y \leq y) &= \Pr\left(1 - \mathrm{e}^{-T} \leq y\right) \\ &= \Pr\left(\mathrm{e}^{-T} \geq 1 - y\right) \\ &= \Pr\left(-T \geq \ln(1-y)\right) \\ &= \Pr\left(T \leq -\ln(1-y)\right) \\ &= F_T\left(-\ln(1-y)\right) \\ &- 1 - \exp\left(-2\left(-\ln(1-y)\right)\right) \\ &= 1 - \exp\left(2\ln(1-y)\right) \\ &= 1 - (1-y)^2 \\ &= 2y - y^2 \end{aligned}$$

where we have made use of the formula for $F_T(t)$ obtained in part (a), the fact that $\lambda = 2$, and that for $0 \leq y < 1$ we have $-2\left(-\frac{\ln(1-y)}{2}\right) > 0$.

So

$$F_Y(y) \;=\; \begin{cases} 0 & y < 0, \\ 2y - y^2 & 0 \le y < 1, \\ 1 & y \ge 1. \end{cases}$$

[5 marks]

Region of non-zero density is $0 < y < 1$. [1 mark]

ii. Within this range,

$$f_Y(y) \;=\; \frac{dF_Y}{dy} = \frac{d}{dy}\left(2y - y^2\right) = 2 - 2y = 2(1-y)$$

as required. [2 marks]

iii. Expectation:

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y)\, dy = \int_0^1 y\,(2(1-y))\; dy \;=\; \int_0^1 2y - 2y^2\, dy$$
$$= \left[y^2 - \frac{2y^3}{3}\right]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$$

[1 mark]

Variance:

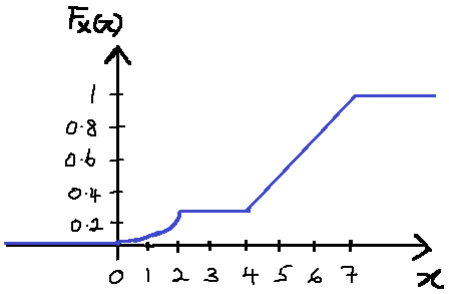
$$E(Y^2) = \int_{-\infty}^{\infty} y^2 f_Y(y)\, dy = \int_0^1 y^2\,(2(1-y))\; dy \;=\; \int_0^1 2y^2 - 2y^3\, dy$$
$$= \left[\frac{2y^3}{3} - \frac{y^4}{2}\right]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6},$$
$$\Rightarrow \mathrm{Var}(Y) = E(Y^2) - (E(Y))^2 \;=\; \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}$$

[2 marks]

[Total 17 Marks]

4. *All parts similar to seen examples.*
CS1 syllabus objective 1.1.3.

(a) Sketch of cdf $F_X(x)$:



[2 marks]

To be a valid cdf, $F_X(x)$ must satisfy: (i) $0 \leq F_X(x) \leq 1$; (ii) $F_X(x)$ is a non-decreasing function; (ii) $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$; (iv) $F_X(x)$ is right-continuous and has left limits. [3 marks]

(b) The random variable X does have a probability density function, because it is clearly continuous and differentiable everywhere except for possibly where the pieces join; at the join points, we have at $x = 0$, $\lim_{x\uparrow 0} 0 = 0 = \frac{x^2}{16}\Big|_{x=0}$; at $x = 2$, $\lim_{x\uparrow 2} \frac{x^2}{16} = \frac{1}{4} = \frac{1}{4}\Big|_{x=2}$; at $x = 4$, $\lim_{x\uparrow 4} \frac{1}{4} = \frac{1}{4} = \frac{x-3}{4}\Big|_{x=4}$; and at $x = 7$, $\lim_{x\uparrow 7} \frac{x-3}{4} = 1 = 1|_{x=7}$. That is, $F_X(x)$ is continuous everywhere, and differentiable everywhere except possibly at up to 3 points. [3 marks]

(c) i. $\Pr(1 < X \leq 5) = F_X(5) - F_X(1) = \frac{2}{4} - \frac{1^2}{16} = \frac{7}{16} = 0.4375$ [1 mark]

ii. $\Pr(X = 4) = 0$, because we know that X has a pdf so is a continuous random variable. [1 mark]

iii. $\Pr(X > 3 \mid X \leq 5) = \frac{\Pr(X>3 \cap X\leq 5)}{\Pr(X\leq 5)} = \frac{\Pr(3<X\leq 5)}{\Pr(X\leq 5)} = \frac{F_X(5)-F_X(3)}{F_X(5)} = \frac{\frac{2}{4}-\frac{1}{4}}{\frac{2}{4}} = \frac{1}{2} = 0.5$ [2 marks]

(d) First need to find pdf $f_X(x)$ of X:

$$f_X(x) = \frac{dF}{dx} = \begin{cases} 0 & x < 0, \\ \frac{x}{8} & 0 \leq x < 2, \\ 0 & 2 \leq x < 4, \\ \frac{1}{4} & 4 \leq x < 7, \\ 0 & x \geq 7. \end{cases}$$

[1 mark]

So with $Y = 2X - 1$,

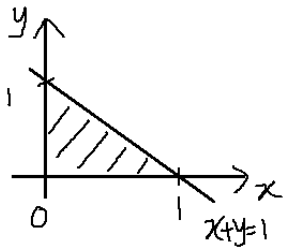
$$\begin{aligned} E(Y) = E(2X-1) = \int_{-\infty}^{\infty} (2x-1) f_X(x)\, dx &= \int_0^2 (2x-1)\frac{x}{8}\, dx + \int_4^7 (2x-1)\frac{1}{4}\, dx \\ &= \int_0^2 \frac{2x^2 - x}{8}\, dx + \int_4^7 \frac{2x-1}{4}\, dx \\ &= \left[\frac{x^3}{12} - \frac{x^2}{16}\right]_0^2 + \left[\frac{x^2}{4} - \frac{x}{4}\right]_4^7 \\ &= \left(\frac{2}{3} - \frac{1}{4}\right) + \left(\frac{49}{4} - \frac{7}{4} - 4 + 1\right) \\ &= \frac{95}{12} \approx 7.9167 \end{aligned}$$

[2 marks]

[Total 15 Marks]

5. *Part (a) similar to seen examples; part (b) unseen.*
CS1 syllabus objectives 1.2.2, 1.2.4, 1.3.1, 1.3.2.

(a) i. Region of non-zero density:



[1 mark]

To find value of c:

$$\begin{aligned}
\iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,dx\,dy &= 1 \\
\Rightarrow \quad \int_{x=0}^{1}\int_{y=0}^{1-x}(cx+1)\,dy\,dx &= 1 \\
\Rightarrow \quad \int_{x=0}^{1}[(cx+1)y]_{y=0}^{1-x}\,dx &= 1 \\
\Rightarrow \quad \int_{x=0}^{1}(cx+1)(1-x)\,dx &= 1 \\
\Rightarrow \quad \int_{x=0}^{1}(1+(c-1)x-cx^2)\,dx &= 1 \\
\Rightarrow \quad \left[x+\frac{(c-1)x^2}{2}-\frac{cx^3}{3}\right]_{x=0}^{1} &= 1 \\
\Rightarrow \quad 1+\frac{c-1}{2}-\frac{c}{3} &= 1 \\
\Rightarrow \quad \frac{c}{6}+\frac{1}{2} &= 1 \\
\Rightarrow \quad c &= 6\left(1-\frac{1}{2}\right)=3
\end{aligned}$$

as required. [2 marks]

ii. From the above sketch, we can see that $\mathrm{Im}X = [0,1]$.
For $0 \le x \le 1$,

$$\begin{aligned}
f_X(x) &= \int_{y\in\mathbb{R}} f_{X,Y}(x,y)\,dy \\
&= \int_{y=0}^{1-x}(3x+1)\,dy \\
&= [(3x+1)y]_{y=0}^{1-x} \\
&= (3x+1)(1-x) = 1+2x-3x^2
\end{aligned}$$

So

$$f_X(x) = \begin{cases} 1+2x-3x^2 & 0\le x\le 1, \\ 0 & \text{otherwise.} \end{cases}$$

[2 marks]

For $0 \le x < 1$, we have

$$\begin{aligned}
f_{Y|X}(y \mid x) &= \frac{f_{X,Y}(x,y)}{f_X(x)} \\
&= \frac{3x+1}{(3x+1)(1-x)} \text{ provided } 0\le y\le 1-x \\
&= \frac{1}{1-x}
\end{aligned}$$

That is, for $0 \leq x < 1$,

$$f_{Y|X}(y \mid x) = \begin{cases} \frac{1}{1-x} & 0 \leq y \leq 1-x, \\ 0 & \text{otherwise,} \end{cases}$$

as required.

[2 marks]

The conditional distribution of Y given $X = x$ is thus the Uniform distribution on the interval $[0, 1-x]$.

[1 mark]

iii. For $0 \leq x < 1$, since $(Y \mid X = x) \sim U(0, 1-x)$, then $E(Y \mid X = x) = \frac{1-x}{2}$.

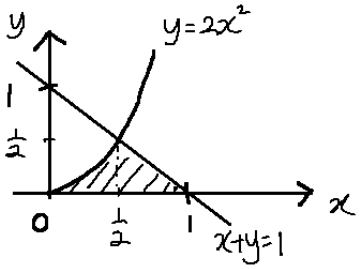
[1 mark]

So now

$$\begin{aligned} E(Y) &= \int_{x \in \mathbb{R}} E(Y \mid X = x) f_X(x)\, dx \\ &= \int_{x=0}^{1} \left(\frac{1-x}{2}\right)\left(1 + 2x - 3x^2\right) dx \\ &= \int_{x=0}^{1} \frac{1 + x - 5x^2 + 3x^3}{2}\, dx \\ &= \left[\frac{x}{2} + \frac{x^2}{4} - \frac{5x^3}{6} + \frac{3x^4}{8}\right]_{x=0}^{1} \\ &= \frac{1}{2} + \frac{1}{4} - \frac{5}{6} + \frac{3}{8} \\ &= \frac{7}{24} \approx 0.2917 \end{aligned}$$

[2 marks]

iv. Region of non-zero density within which $y < 2x^2$:



[1 mark]

So required probability is

$$\begin{aligned}
\Pr(Y < 2X^2) &= \int_{y=0}^{1/2} \int_{x=\sqrt{y/2}}^{1-y} (3x+1)\,dx\,dy \\
&= \int_{y=0}^{1/2} \left[\frac{3x^2}{2} + x\right]_{x=\sqrt{y/2}}^{1-y} dy \\
&= \int_{y=0}^{1/2} \left(\frac{3(1-y)^2}{2} + (1-y) - \frac{3y}{4} - \sqrt{y/2}\right) dy \\
&= \int_{y=0}^{1/2} \left(\frac{3-6y+3y^2}{2} + (1-y) - \frac{3y}{4} - \sqrt{y/2}\right) dy \\
&= \int_{y=0}^{1/2} \left(\frac{5}{2} - \frac{19y}{4} + \frac{3y^2}{2} - \sqrt{y/2}\right) dy \\
&= \left[\frac{5y}{2} - \frac{19y^2}{8} + \frac{y^3}{2} - \frac{2y^{(3/2)}}{3\sqrt{2}}\right]_{y=0}^{1/2} \\
&= \frac{5}{4} - \frac{19}{32} + \frac{1}{16} - \frac{1}{6} \\
&= \frac{53}{96} \approx 0.5521
\end{aligned}$$

[3 marks]

(b) For independent U, V,

$$\begin{aligned}
\mathrm{Cov}(U, UV) &= E(U(UV)) - E(U)E(UV) \\
&= E(U^2)E(V) - (E(U))^2 E(V) \text{ (using independence)} \\
&= E(V)\left(E(U^2) - (E(U))^2\right) \\
&= E(V)\mathrm{Var}(U)
\end{aligned}$$

as required. [3 marks]

[Total 18 Marks]

6. *Parts (a) and (b) similar to seen examples; part (c) unseen.*
CS1 syllabus objectives 1.1.1, 1.2.6, 1.5.1.

(a) $E(X_i) = \frac{1}{6}(1+2+3+4+5+6) = \frac{7}{2} = 3.5$ [1 mark]

To find variance:

$$\begin{aligned}
E(X_i^2) &= \frac{1}{6}\left(1^2+2^2+3^2+4^2+5^2+6^2\right) = \frac{91}{6} \\
\Rightarrow \quad \mathrm{Var}(X_i) &= \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12} \approx 2.9167
\end{aligned}$$

[2 marks]

(b) With $n = 100$, then

$$\begin{aligned} E(S_{100}) &= 100E(X_i) = 350, \\ \text{Var}(S_{100}) &= 100\text{Var}(X_i) = \frac{3500}{12} = \frac{875}{3} \end{aligned}$$

[2 marks]

By CLT, since $n = 100$ is large, the distribution of S_{100} can be approximated by the $N\left(350, \frac{875}{3}\right)$ distribution. Note also that S_{100} is integer-valued, so a continuity correction is appropriate.

Hence

$$\begin{aligned} \Pr(S_{100} \geq 380) &= \Pr(S_{100} \geq 379.5) \text{ (continuity correction)} \\ &= \Pr\left(\frac{S_{100} - 350}{\sqrt{875/3}} \geq \frac{379.5 - 350}{\sqrt{875/3}}\right) \\ &\approx \Pr(Z \geq 1.73) \text{ where } Z \sim N(0,1) \\ &= 1 - \Phi(1.73) = 1 - 0.9582 \text{ (from tables)} \\ &= 0.0418 \end{aligned}$$

[4 marks]

(c) With $n = 6$, then $E(S_6) = 6 \times 3.5 = 21$ and $\text{Var}(S_6) = 6 \times \frac{35}{12} = \frac{35}{2}$. [1 mark]

Now

$$\begin{aligned} \Pr(S_6 \geq 50) &\leq \Pr(|S_6 - 21| \geq 29) \\ &\leq \frac{35/2}{29^2} = \frac{35}{1682} \approx 0.0208. \end{aligned}$$

[3 marks]

(Note: In fact, $\text{Im}(S_6) = \{6, 7, \ldots, 36\}$, so that $\Pr(S_6 \geq 50) = 0$, and the Chebyshev bound is not very useful here.)